Model of Models: When Does Emitting a Specialist Beat Attending, Adapting, or Tuning?

A Controlled Cross-Domain Study of Amortized Weight Generation

John C. Howell*

Nineoneofour

June 11, 2026

Abstract

We show that the weights a hypernetwork emits for a task-specific specialist are *algebraically composable*: linearly interpolating the weight vectors of two emitted specialists yields a coherent blend of their functions—a circle-to-square morph from θ_{circle} and θ_{square}—a capability no in-context or gradient-adaptation method possesses. We situate this within a broader question: given a task described by a few examples, how should a model be specialized to it? Four mechanisms are available—zero-shot, in-context attention, test-time gradient adaptation, and emitting specialist weights—yet the operating regime of the last is rarely mapped. We run the identical four-way comparison across six tasks spanning regression, generation, language modeling, reinforcement learning, and clinical and genomic classification, under matched budgets. Emitting specialists dominates low-dimensional, amortizable task families: 380× lower error than MAML on few-shot sinusoid regression at *zero* test-time gradient steps, noise-floor shape generation at 840× lower inference cost, and parity with the state-of-the-art amortized tabular model (TabPFN) on clinical few-shot classification while emitting a *reusable* specialist rather than re-attending the support set per query. It cannot match in-context attention on high-dimensional sequence modeling, and the gap *widens with scale*: a one-pass adapter captures only 16% then 11% of the in-context gain as the base model grows from 5M to 15M parameters. Mechanism ablations confirm the emitted specialist is genuinely task-conditioned. We close with a falsifiable thesis bounding when each conditioning mechanism should be preferred.

1 Introduction

A recurring problem in machine learning is to specialize a general model to a specific task that is described only by a handful of examples: a few labeled patients, a few input–output grids, a few transitions from a new environment. Four mechanisms are in common use. *Zero-shot* ignores the examples. *In-context* learning prepends them and lets attention condition the forward pass. *Gradient adaptation*—MAML and ordinary fine-tuning—takes a few optimization steps on the examples. The fourth, which we study here, *emits* the parameters of a small task-specific specialist network in a single forward pass of a hypernetwork. The mechanism is not new: hypernetworks, conditional neural processes, and context-to-LoRA generators all instantiate it. What is missing is a map of *when* it should be preferred over the other three.

We provide that map. We define a single formulation in which all four mechanisms share a backbone and a budget (§3, Fig. 1) and run the identical comparison across six tasks spanning

*Nineoneofour is the operating name of Project[LNCLN]. Correspondence: `jc@nineonefour.org`.

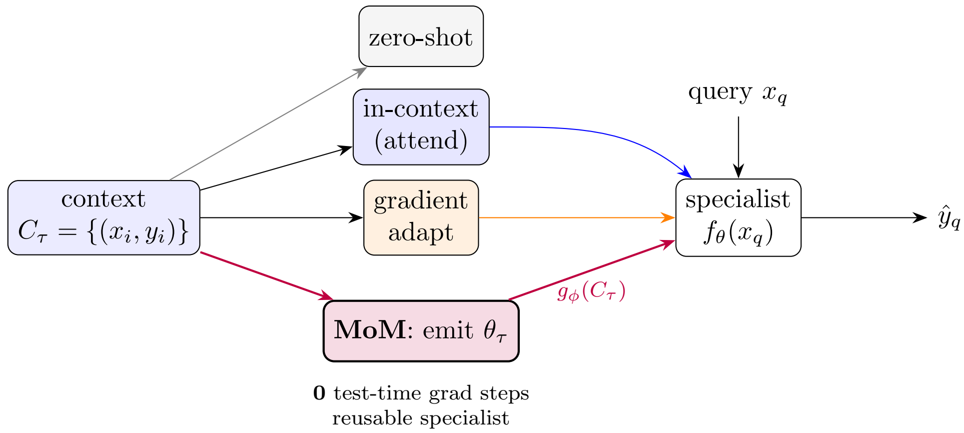

Figure 1: The four conditioning mechanisms over a shared specialist f_θ and query x_q. Zero-shot ignores the context; in-context attends to it every query; gradient adaptation takes SGD steps on it; **MoM** emits the specialist's weights $\theta_\tau = g_\phi(C_\tau)$ in one pass, with no test-time gradient steps and no context in the query window.

five domains (§4): few-shot sinusoid regression, parametric shape generation, byte-level language modeling, meta-reinforcement learning, and clinical and genomic few-shot classification. Rather than reporting only where emission wins, we deliberately chart where it *ties* and where it *loses*, and we measure how those boundaries move with scale.

Our most striking positive result is that emitted specialists are **algebraically composable**: linearly interpolating the weight vectors of two emitted specialists produces a coherent blend of their functions (§5.2, Fig. 3)—a property no in-context or gradient-adaptation method exhibits, and one that suggests the hypernetwork learns an approximately linear map from task to weight space. Our most consequential negative result is a scaling trend: on language modeling, a one-pass emitted adapter captures a *shrinking* fraction of the in-context performance gain as the base model grows (§5.3). Together they bound the method's regime.

Contributions.

(C1) A controlled four-way comparison of conditioning mechanisms across six tasks and five domains under matched budgets (§4).

(C2) **Boundary maps** rather than wins: where emission dominates, ties, and loses—including a scaling trend in which the gap to in-context attention *widens* for high-dimensional sequences (§5.3).

(C3) **Mechanism ablations** establishing that the emitted specialist is genuinely task-conditioned, not a memorized prior (§5.1).

(C4) **Weight-space composition**: emitted specialists interpolate and blend algebraically, including on a function family not closed under addition (§5.2).

Thesis (falsifiable). *Amortized weight-emission dominates within a trained low-dimensional task family, at far lower test-time cost than gradient adaptation or in-context attention, but it cannot match in-context attention for high-dimensional sequence modeling, and that gap grows with model scale.*

2 Related Work

Generating weights. Hypernetworks (Ha et al., 2017) introduced networks that output the parameters of another network. Conditional Neural Processes (Garnelo et al., 2018) and functa amortize this over a task distribution, emitting an instance-specific function from a context set—the formulation our shape and regression experiments instantiate. We adopt FiLM-style modulation (Perez et al., 2018) as a low-dimensional alternative to emitting raw weights.

Context-to-PEFT for language. A line of work generates parameter-efficient adapters from context: HyperTuning and HyperT5 emit soft prefixes and LoRA weights from few-shot demonstrations (Phang et al., 2023; Phang, 2024), and more recent systems map context to LoRA in a single pass (Liu et al., 2026; Lv et al., 2024). Our language experiments use the same recipe; our contribution there is the controlled comparison against in-context attention and the scaling trend (§5.3), not the adapter-generation mechanism.

Meta-learning and meta-RL. Gradient-based meta-learning (MAML (Finn et al., 2017), CAVIA (Zintgraf et al., 2019)) adapts via test-time SGD; context-based meta-RL (PEARL (Rakelly et al., 2019)) conditions on an inferred latent. Beck et al. (2023) show hypernetwork-generated policies are competitive with latent conditioning in meta-RL—a finding our RL experiment reproduces and situates within the broader comparison.

Amortized tabular inference. TabPFN (Hollmann et al., 2023) is a pretrained transformer that classifies tabular data in-context in one forward pass; it is the strongest baseline for our few-shot tabular tasks, and we compare against it directly.

Decoupled training. The appendix relates a negative result to synthetic gradients / DNI (Jaderberg et al., 2017) and decoupled greedy local learning (Nøkland and Eidnes, 2019; Belilovsky et al., 2019).

Positioning. We do not claim the weight-emission mechanism is new. Our contribution is a controlled cross-domain map of *when* it should be preferred over the three alternatives, a scaling characterization of its principal failure mode, and the observation that emitted specialists compose in weight space.

3 Method: The MoM Quadrant

Task and context. A task τ is described by a context set $C_\tau = \{(x_i, y_i)\}_{i=1}^{K}$ of K labeled examples; methods are evaluated on held-out queries x_q drawn from the same τ. We hold K, the query distribution, and the specialist architecture fixed across all four mechanisms so that differences reflect the conditioning mechanism rather than capacity or data.

The specialist and its emitter. The specialist f_θ is a small network (an MLP or a few-layer transformer, depending on the domain). An encoder g_ϕ maps the context set to specialist parameters, $\theta_\tau = g_\phi(C_\tau)$, and is trained end-to-end by differentiating the query loss through the generated weights. We study two emission variants: a *direct* variant in which g_ϕ outputs the full weight vector θ_τ, and a *FiLM* variant in which it outputs per-layer scale and shift modulations of a shared base specialist—trading expressiveness for far fewer emitted values. The encoder is permutation-invariant in the context examples (a deep-sets pool or a set transformer). Two implementation details proved necessary: emitting weights against RMS-normalized targets (raw target magnitudes span orders of magnitude), and matching the specialist's input featurization to the data's function class (§4).

The four mechanisms. Over this shared specialist (Fig. 1): *zero-shot* uses fixed θ_0 and ignores C_τ; *in-context* prepends C_τ to the query and conditions through attention with no weight change; *gradient adaptation* produces θ_τ by n SGD steps on C_τ starting from a meta-learned θ_0 (MAML) or from scratch (per-task fine-tune); and *MoM* sets $\theta_\tau = g_\phi(C_\tau)$ in a single forward pass.

Why emission can be cheaper at deployment. MoM takes *zero* test-time gradient steps, unlike adaptation; the emitted specialist is typically orders of magnitude smaller than the backbone and is *reused* across all queries for a task, unlike in-context attention, which re-processes the context for every query and pays a context-length KV cache. These structural savings hold even when accuracy is at parity, and are the lens through which we read the results.

4 Experiments

We organize the six tasks from the most amortizable (low-dimensional task latent) to the least. For each we give the protocol, the four-way comparison, and a verdict, leading with where emission wins and stating plainly where it does not. All MoM and baseline models are trained under matched budgets; where a published protocol exists we reproduce its baselines to calibrate our harness.

4.1 Sinusoid few-shot regression (low-dim, amortizable)

We follow the MAML/CAVIA protocol: each task is $y = A\sin(x - \varphi)$ with $A \sim U[0.1, 5]$, $\varphi \sim U[0, \pi]$, inputs in $[-5, 5]$, K=10 context points, and MSE over 100 held-out queries; the specialist is the standard 1–40–40–1 MLP used by all methods. Our protocol-exact MAML reproduction (0.171 MSE) lands in the published range (0.19–0.29), calibrating the harness.

Method	MSE ↓	test-time grad steps
CAVIA (published)	0.19–0.21	5
MAML (published)	0.23–0.29	5
MAML (our repro, protocol-exact)	0.171 ± 0.064	5
MoM-direct	$\mathbf{0.0004 \pm 0.0004}$	**0**
MoM-FiLM	$\mathbf{0.0005 \pm 0.0001}$	**0**

Table 1: Sinusoid 10-shot (5 seeds). MoM ≈380× lower MSE at zero adaptation steps. OOD (§5.3): amplitude shift — MoM graceful, MAML diverges; frequency shift — both fail, MAML 2× more robust.

4.2 Vector-shape generation (low-dim, amortizable)

Each task is a parametric 2D curve (circle, square, spiral, etc.) under random rotation, scale, and noise, observed as a sequence of points. The baseline is an autoregressive transformer that rolls out the curve point by point; MoM emits a specialist $t \mapsto (x, y)$ from a few context points and renders the whole curve in parallel. Accuracy is comparable but the resource gap is the point.

Method	continuation MSE ↓	gen time (256 shapes)	per-instance size
AR transformer (ours)	0.0041	573 ms	202k params
MoM hyper→specialist	**0.0018** (noise floor)	**21 ms**	1.7k params
specialist alone (amortized)	0.0018	**0.68 ms**	**132 floats**

Table 2: MoM beats the AR baseline on accuracy (2.3×, at the noise floor) and on cost (26–840× faster, ∼1500× smaller program). *Gotcha for appendix: specialist function-class must match data — a periodic Fourier basis fails on open curves; diagnose by direct per-instance fitting.*

4.3 Few-shot tabular: clinical & genomic identification

We use two real datasets: the UCI HCV liver panel (12 blood markers; healthy vs. liver disease) and the TCGA PAN-CAN RNA-seq corpus (top-1000 high-variance genes; 5 tumor types). On the full data our logistic-regression and random-forest references match published SOTA (0.99 AUC; 0.999 accuracy), confirming the data is solved and calibrating the harness. The MoM-relevant regime is therefore few-shot: a task is a small balanced support set, and we compare emission against in-context attention, per-task gradient fitting, and the state-of-the-art amortized tabular model TabPFN.

	Blood (HCV), 16-shot		Genes (TCGA), 5-shot
Method	ROC-AUC ↑	infer cost	accuracy ↑
Full-data SOTA (ref)	0.989–0.994	—	0.999
TabPFN (SOTA amortized)	0.967	601 ms/task, re-attend/query	0.991†
MoM (emit classifier)	**0.966**	reusable specialist, ∼ms	0.998
in-context attention	0.925	full support/query	1.000
per-task logistic fit	0.930	grad steps/task	0.996

Table 3: MoM **ties SOTA TabPFN** on blood (0.966 vs 0.967) while beating gradient-tuning and in-context — and emits a *reusable* specialist whereas TabPFN re-attends the full support set for every query. Genes is a saturated tie (near-separable). †TabPFN capped at PCA-20 by its feature limit; MoM uses all 1000 genes.

4.4 Meta-RL: HalfCheetah-Vel (PEARL protocol)

Tasks are target running velocities; an episode-1 trajectory is the context for episode-2 control. We hold the SAC machinery and a context encoder fixed and vary only the conditioning mechanism: *latent* concatenates the inferred task embedding to the policy input (PEARL-style), while *MoM* uses it to FiLM-modulate the policy's weights. We report the standard best-checkpoint return and the final return over 3 seeds.

4.5 Language modeling (high-dim): enwik8 byte LM

The backbone is a frozen byte-level transformer pretrained on enwik8. MoM's hyper-LoRA reads 512 context bytes and emits a rank-4 LoRA adapter; the model then predicts a contiguous following chunk with no context in its own window. We compare against zero-shot, in-context (the 512 bytes prepended), and a per-document Adam-tuned LoRA. This is a high-dimensional task latent (the "style" of an arbitrary document), and emission does not win.

Method	best-ckpt return ↑	final return	conditioning
PEARL (published)	≈ -65	—	latent concat
latent (our PEARL-style)	-50 ± 15	-48 ± 11	concat
MoM (FiLM policy)	$\mathbf{-57 \pm 5}$	-70 ± 11	emit weights

Table 4: 3 seeds. On the standard best-checkpoint metric, latent and MoM are a **statistical tie** (overlapping CIs); MoM is markedly **more consistent** (std 5 vs 15). Latent edges MoM on final return. Both reach/beat published PEARL. Consistent with Beck et al. (2023): hypernet policies competitive, not dominant.

Method	bits/char ↓	context tokens in window
zero-shot (frozen 5M base)	1.814	0
in-context (512 prepended)	**1.645**	512
MoM hyper-LoRA	1.789	**0**
per-doc Adam-tuned LoRA	2.557	0

Table 5: MoM captures only 15% of the in-context gain (but beats gradient tuning, which is *worse than zero-shot*). See §5.3 for the scaling trend.

4.6 Puzzle solving: ARC (high-dim, compositional)

We train on procedurally generated RE-ARC tasks: MoM reads three input–output grid pairs and emits a specialist that transforms a held-out test grid. We evaluate on held-out instances of seen rules and on the real ARC training and evaluation sets.

- Held-out generated (seen rules, new instances): **11.5%** exact match.
- Real ARC training grids: 3.6%. Real ARC **evaluation (unseen rules): 0.0%**.
- **Verdict:** clean negative. Amortization *interpolates the trained task family* but does not *synthesize new programs*; ARC SOTA (40–55%) uses search/LLMs far outside our budget.

5 Analysis

5.1 Mechanism ablations: is the specialist really task-conditioned?

A natural objection is that the encoder ignores its context and the system merely memorizes the task distribution. Three ablations on the sinusoid MoM refute this.

- **Context-shuffle:** matched-context MSE 0.0003 vs wrong-task-context 5.97 (∼19,000× worse, and worse than the 4.14 target variance — confidently wrong, not a memorized prior).
- **Specialist-swap:** task-A specialist on A 0.0003 vs B-specialist on A 5.96. Each emitted network is genuinely task-specific.
- **Context-size sweep:** K=1:1.84, 2:0.31, 3:0.06, 5:0.004, 10:0.0003 — monotonic, resolves as K passes the 2 latent params. Inference, not lookup (Fig. 2b).

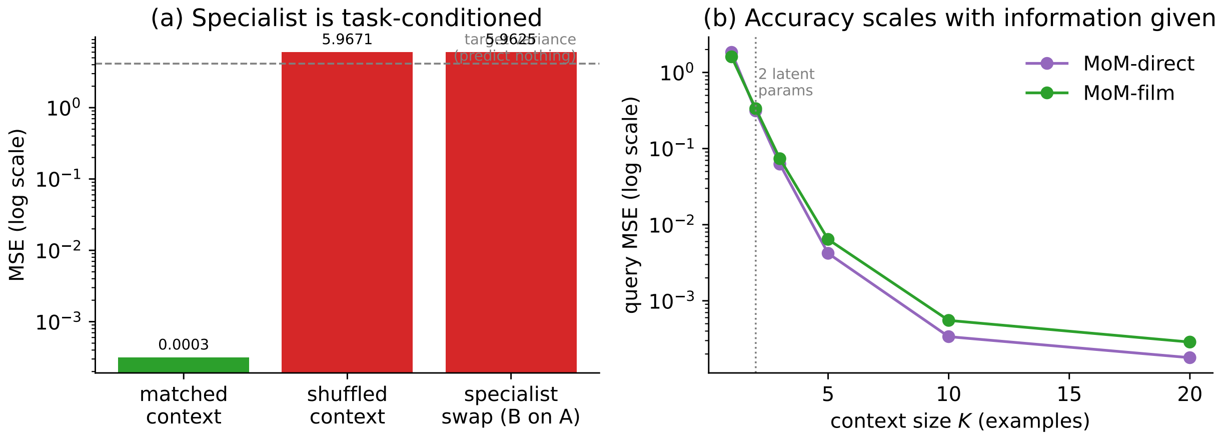

Figure 2: Mechanism ablations on the sinusoid MoM. (a) Replacing the context with another task's examples, or running task A's specialist on task B, raises error ∼19,000×—past the variance of predicting nothing—so the specialist is fully context-determined. (b) Query error falls monotonically with context size and resolves once K exceeds the two latent task parameters: the encoder performs inference, not retrieval.

5.2 Weight-space composition (candidate novelty)

If the encoder maps tasks to weights approximately linearly, emitted specialists should compose. They do.

- Interpolating two emitted specialists' weights $\theta = (1-\alpha)\theta_A + \alpha\theta_B$ tracks the blended target: **sinusoid** midpoint MSE 0.071 vs 6.03 task distance (85× below); **shapes** (a family *not* closed under addition) 0.032 vs 0.695 (22× below) — coherent circle→square morph (Fig. 3).
- Emitted weight-space is approximately linear in the task latent ⇒ specialists compose algebraically; no in-context/MAML/PEARL baseline can interpolate task programs.
- **[TODO: extend: arithmetic (A−B+C); generalization to unseen combinations.]**

`exp22b_morph.png` — weight interpolation $\theta_{\text{circle}} \rightarrow \theta_{\text{square}}$
(purple = weight-blend output, gray = geometric target)

Figure 3: Specialists compose in weight space, even for a non-closed family.

5.3 Scaling: does the in-context gap close?

One might hope a one-pass adapter merely lags in-context attention at small scale and catches up as the backbone grows. The opposite holds. As the enwik8 base grows from 5M to 15M parameters

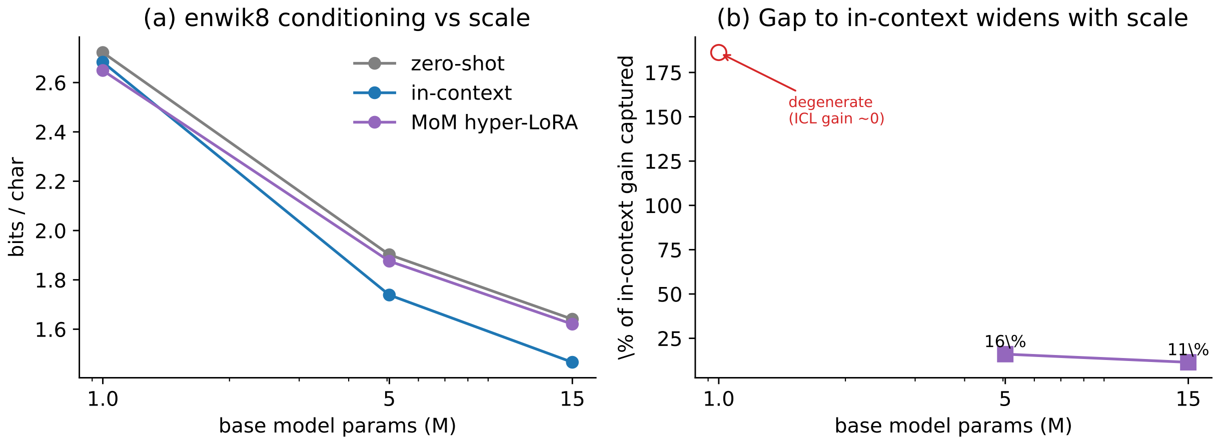

Figure 4: enwik8 scaling. (a) All three conditioning curves improve with base size, but the MoM–in-context gap does not close. (b) The fraction of the in-context gain captured by a one-pass emitted adapter shrinks from 16% to 11% over $5 \to 15$M params; the 1M point is degenerate (in-context gain ≈ 0).

the in-context gain itself grows ($0.163 \to 0.175$ bits/char), but the fraction MoM recovers *shrinks* ($16\% \to 11\%$; Table 5.3, Fig. 4). The 1M point is degenerate—in-context barely helps so weak a model, making the ratio uninformative. The trend gives the thesis its upper boundary: for high-dimensional sequence modeling, emission does not close the gap to attention, and the gap widens with scale.

base size	zero-shot	in-context	hyper-LoRA	% ICL gain captured
1.0M	2.722	2.683	2.649	186%‡
5.1M	1.902	1.739	1.876	16%
14.7M	1.640	1.466	1.620	11%

Table 6: enwik8. From 5M→15M the in-context gain grows (0.163→0.175) while MoM's capture *shrinks* (16%→11%): the gap widens with scale. ‡1M degenerate (ICL gain only 0.039, ratio uninformative).

6 Limitations

Our experiments are at small scale—5–15M-parameter language models, MLP and few-layer-transformer specialists, two consumer GPUs. We measure a scaling *trend*, but the absolute regime is small, and the language and ARC conclusions should be read as bounds at this scale rather than at frontier scale. Emission fails on high-dimensional and compositional tasks (language modeling, ARC), and on language the failure *grows* with scale, so the method is not a general-purpose substitute for attention. Out of distribution, amortization extrapolates gracefully under scaling shifts but breaks under structural shifts and is confidently wrong rather than abstaining—a safety-relevant failure mode for deployment. Finally, weight-space composition is demonstrated on synthetic families; whether emitted specialists for real tasks compose as cleanly is open.

7 Conclusion

Across six tasks and five domains we find a consistent pattern: emitting a task-specific specialist dominates when the task family is low-dimensional and amortizable—matching or beating gradient adaptation and the strongest amortized baselines at far lower test-time cost—but cannot match in-context attention for high-dimensional sequence modeling, where the gap widens with scale. The practical decision rule that follows is simple: *emit a specialist when the task family is narrow and deployment cost matters; attend in-context when the task latent is rich, especially at scale.* Two further findings sharpen the picture: mechanism ablations show the emitted specialist is genuinely task-determined, and emitted specialists compose algebraically in weight space—a property absent from the alternatives and a promising direction for turning amortized emission from a conditioning trick into a manipulable representation of tasks.

A Negative result: predicted gradients vs. local losses

This project began with a different question—whether a transformer could be trained without global backpropagation by having each layer *predict* its own incoming gradient (a decoupled-neural-interfaces / synthetic-gradient scheme (Jaderberg et al., 2017))—and the negative result that emerged is worth recording, both because it is clean and because it motivated the amortization view that became MoM.

On a shape-classification task, per-token gradient predictors with RMS-normalized targets and a bootstrapped target chain do train a small transformer with zero global backward passes, reaching 96–99.6% accuracy at two layers. But the approach has two limits. First, it does not scale in depth: bootstrap error compounds and accuracy collapses beyond a few layers *unless* each stage is given an auxiliary local loss that supplies an exact gradient anchor. Second—and this is the negative result—once those local anchors are present, the gradient predictors are *redundant*: an ablation that removes them and keeps only the per-stage auxiliary losses (greedy local learning, Nøkland and Eidnes, 2019; Belilovsky et al., 2019) matches end-to-end backpropagation (100% / 99.8% at 4 / 6 layers on classification; 0.0033 vs. 0.0022 MSE on autoregressive generation) while holding peak training memory nearly flat in depth (740 MB vs. backpropagation's 6927 MB at 24 layers, a 9.4× reduction).

A useful diagnostic emerged along the way: *gradient predictability decays over training.* The cosine similarity between a layer's true and predictable gradient is ≈ 0.99 at initialization, 0.6–0.8 at mid-training, and near zero at convergence—gradients become noise as the loss flattens. This bounds any synthetic-gradient method to the early, high-loss regime and explains why such methods help least exactly when one most wants a shortcut. The broader lesson—that *amortizing* a learned signal beats predicting a moving target—carried directly into the specialist-emission framing of the main paper. This appendix could stand as a short negative-results contribution on its own.

B Reproducibility

All datasets are public: UCI HCV, UCI TCGA PAN-CAN RNA-seq, enwik8, ARC-AGI with the RE-ARC procedural generators, and Gymnasium HalfCheetah-v5. All models are implemented in PyTorch and trained on consumer GPUs (2×NVIDIA A2 and 2×RTX 5070). Code and configurations: `https://github.com/johnchowell/Model-of-Models`. Per-experiment settings are in Table 7; results report multiple seeds where noted in §4 (sinusoid: 5; meta-RL: 3; tabular: 5-fold cross-validation).

Experiment	Key settings
Sinusoid (§5.1)	specialist MLP 1–40–40–1; encoder deep-sets, d=128; MoM 30k steps, batch 256, Adam lr 10^{-3} cosine; MAML 70k meta-steps, batch 25, inner lr 0.01 (1 step), outer lr 10^{-3}, 2nd-order; K=10.
Shapes (§5.2)	specialist Fourier(t)+t → MLP, 1.7k params (FiLM: 132 mods); encoder transformer d=128; 8k steps, batch 256, lr 10^{-3}; context 12 points, curve length 32.
Tabular (§5.3)	few-shot net d=128 (genes) / row-encoder + FiLM head; 3–4k steps, batch 32–64, lr 10^{-3}, wd 10^{-4}; blood K=16 balanced, genes 5-way 5-shot, top-1000 genes; TabPFN v2 (PCA-20 for genes).
Meta-RL (§5.4)	SAC, policy/critic hidden 256; 300 iters, batch 256, lr 3×10^{-4}, γ=0.99, τ=5×10^{-3}; context encoder $z\in\mathbb{R}^{16}$, 64 context transitions; episodes 200 steps.
enwik8 (§5.5)	base byte-LM d=256, 6 layers, T=768, pretrain 8k steps, batch 32, lr 3×10^{-4} OneCycle; hyper-LoRA rank 4 on $\{o, \mathrm{fc}_2\}$, 4k steps; context 512 bytes, target 256.
ARC (§5.6)	grid transformer, enc 4 / dec 6 layers, d=256, 12×12 canvas; 60k steps, batch 32, lr 3×10^{-4}; 3 example pairs; trained on RE-ARC, grids $\leq$ 12×12.
Scaling (§6.3)	bases $(d, L) \in \{(128, 4), (256, 6), (384, 8)\}$; pretrain 6k, hyper-LoRA 3k steps each.

Table 7: Per-experiment configurations.

References

Jacob Beck, Matthew Jackson, Risto Vuorio, and Shimon Whiteson. Hypernetworks in meta-reinforcement learning. *Conference on Robot Learning (CoRL)*, 2023.

Eugene Belilovsky, Michael Eickenberg, and Edouard Oyallon. Greedy layerwise learning can scale to ImageNet. In *International Conference on Machine Learning (ICML)*, 2019.

Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning (ICML)*, 2017.

Marta Garnelo, Dan Rosenbaum, Chris J. Maddison, Tiago Ramalho, David Saxton, Murray Shanahan, Yee Whye Teh, Danilo J. Rezende, and S. M. Ali Eslami. Conditional neural processes. In *International Conference on Machine Learning (ICML)*, 2018.

David Ha, Andrew Dai, and Quoc V. Le. Hypernetworks. In *International Conference on Learning Representations (ICLR)*, 2017.

Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations (ICLR)*, 2023.

Max Jaderberg, Wojciech Marian Czarnecki, Simon Osindero, Oriol Vinyals, Alex Graves, David Silver, and Koray Kavukcuoglu. Decoupled neural interfaces using synthetic gradients. *International Conference on Machine Learning (ICML)*, 2017.

Yewei Liu, Xiyuan Wang, Yansheng Mao, Yoav Gelberg, Haggai Maron, and Muhan Zhang. SHINE: A scalable in-context hypernetwork for mapping context to LoRA in a single pass. *arXiv preprint arXiv:2602.06358*, 2026.

Chuancheng Lv, Lei Li, Shitou Zhang, Gang Chen, Fanchao Qi, Ningyu Zhang, and Hai-Tao Zheng. HyperLoRA: Efficient cross-task generalization via constrained low-rank adapters generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16376–16393, 2024.

Arild Nøkland and Lars Hiller Eidnes. Training neural networks with local error signals. In *International Conference on Machine Learning (ICML)*, 2019.

Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *AAAI Conference on Artificial Intelligence*, 2018.

Jason Phang. Investigating the effectiveness of hypertuning via gisting. *arXiv preprint arXiv:2402.16817*, 2024.

Jason Phang, Yi Mao, Pengcheng He, and Weizhu Chen. Hypertuning: Toward adapting large language models without back-propagation. In *International Conference on Machine Learning (ICML)*, 2023.

Kate Rakelly, Aurick Zhou, Chelsea Finn, Sergey Levine, and Deirdre Quillen. Efficient off-policy meta-reinforcement learning via probabilistic context variables. In *International Conference on Machine Learning (ICML)*, 2019.

Luisa Zintgraf, Kyriacos Shiarlis, Vitaly Kurin, Katja Hofmann, and Shimon Whiteson. Fast context adaptation via meta-learning. In *International Conference on Machine Learning (ICML)*, 2019.